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Towards a new method of generating predictive and interpretable machine learning models and its application to the carbon capture process system

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Abstract

Machine Learning (ML) models have demonstrated outstanding performance in predicting essential parameters in the carbon capture process system and support a better understanding of the relationships among the parameters. Their effectiveness in accurately processing and analyzing large volumes of data is well-established. However, these models often function as “black boxes,” and the reasoning or processes of the models are often unknown. Practitioners often struggle to understand how specific inputs result in particular outputs. This lack of transparency is a barrier to the broader adoption of ML approaches in sectors such as healthcare, finance, heavy industry, and law, where decisions often need to be transparent and justifiable.

Therefore, increasing transparency in ML models is essential for enhancing the adoption of the ML approach. One possible solution is to incorporate Explainable Artificial Intelligence (XAI) techniques, which aim to clarify the decision-making processes of the models. For instance, feature importance metrics and attention mechanisms can identify the most critical inputs in decision-making. By shedding light on the inner mechanisms of ML models, practitioners can enhance their understanding and build confidence in using ML technologies.

This paper presents a method designed to enhance the transparency of ML models. Our approach uses advanced modelling techniques and model explanation methods, namely the Decision Trees Ensemble (DTE) and Tree-Based Local Interpretable Model-agnostic Explanation (LIMEtree), to make predictions more understandable for practitioners. To validate the approach, a new dataset was generated from a ProMax simulation, in which specifications for the contactors are derived from existing carbon dioxide capture units in North America. Using the same method Wang et al. proposed [1], we first developed an accurate correlation model of the relationships of parameters in the carbon capture process system with the help of a DTE, GAN (Generative Adversarial Network) and PFA (Principal Feature Analysis). Then, we applied the LIMEtree method to interpret the DTE model's predictions.

Keywords: Post Combustion Carbon Capture; Machine Learning; Explainable Artificial Intelligence; Tree-Based Local Interpretable Model-agnostic Explanation; Decision Trees Ensemble; Post Combustion; CO₂ Capture; Interpretability

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1. Introduction

Over the past twenty years, researchers have made significant progress in utilizing Machine Learning (ML) techniques to enhance the Post-Combustion Carbon Capture (PCC) process. They have created various models, from basic linear regression to complex deep Neural Networks (NN)[1-6]. Many ML models demonstrate strong performance but are seldom adopted in industrial applications. This cautious approach is justified because making decisions based on complex black-box ML models that surpass human understanding is hazardous. Doing so also raises important ethical considerations regarding accountability [7], safety [8], and liability [9].

Investigating the correlations among key parameters in the PCC process is essential for optimizing these systems [5]. This investigation focuses not only on the accuracy of predictions but also on the explainability and transparency of the ML models, both considerations are crucial for adoption by industrial practitioners [10, 11]. We propose using the Tree-Based Local Interpretable Model-agnostic Explanation (LIMEtree) approach to analyze the correlations among PCC process parameters. LIMEtree offers consistent, high-fidelity, and measurable explanations for each prediction made by the accurate Decision Tree Ensembles (DTE) models. These explanations enhance practitioners' trust in the models' predictions.

The next section of the introduction will discuss a literature review of the methods of GAN (Generative Adversarial Network), PFA (Principal Feature Analysis), DTE (Decision Tree Ensembles) and LIME (Local Interpretable Model-agnostic Explanation) and conclude with the research objectives and contributions.

Nomenclature			
CCE	Carbon Capture Efficiency	LIME	Local Interpretable Model-agnostic Explanation
C-GAN	Conditional Generative Adversarial Network	LIMEtree	Tree-Based Local Interpretable Model-agnostic Explanation
CSI	Coefficient Stability Index	ML	Machine Learning
CT-GAN	Conditional Tabular Generative Adversarial Network	PCA	Principal component analysis
DF	Decision Forest	PFA	Principal Feature Analysis
DT	Decision Tree	PCC	Post Combustion Carbon Capture
DPoI	Data Point of Interest	QoE	Quality of Explanation: CSI, VSI and R-Squared
DTE	Decision Tree Ensembles	RBF	Radial Basis Function
GAN	Generative Adversarial Network	VSI	Variables Stability Index
GBGT	Stochastic Gradient Boosted Decision Trees	XAI	Explainable Artificial Intelligence

1.1. GAN, CGAN and CTGANs

A Generative Adversarial Network (GAN), introduced by Goodfellow et al.[12], is a machine learning model that generates new data resembling a given dataset using two neural networks: a generator and a discriminator. The generator creates synthetic data while the discriminator assesses its authenticity. Through an adversarial process, the generator improves at producing convincing data, while the discriminator becomes better at distinguishing between real and fake samples, ultimately enabling the generation of realistic data.

A Conditional Generative Adversarial Network (C-GAN) [13] extends GAN by adding conditioning that guides the generation process with specific constraints. Like GAN, C-GAN consists of a generator and a discriminator, but the generator is conditioned to create data that fits certain criteria while retaining the original data's characteristics.

While GANs and C-GANs have been used to generate text and images, their application to tabular data has been less explored. The CT-GAN model [14] addresses this gap, efficiently generating both continuous and categorical data. Key innovations include mode-specific normalization, a conditional generator, and training-by-sampling, which allow CT-GAN to model complex feature distributions.

1.2. PCA and PFA

Principal Component Analysis (PCA) reduces data dimensionality by generating principal components to present high-dimensional data [15]. The process normalizes the data, computes the covariance matrix, and performs eigenvalue decomposition to transform the data into uncorrelated principal components that capture the most variance. To select the most influential features, the Principal Feature Analysis (PFA) [16] applies PCA and uses K-mean to cluster eigenvalues [17]. It identifies the most influential features by finding the features closest to the cluster centers.

1.3. DTE

Decision Tree Ensembles (DTEs) leverage decision trees [18] as weak learners, forming a robust model for tabular data [19, 20]. With minimal preprocessing, DTEs can handle both numerical and categorical features. As weak learners, decision trees split data recursively, creating hyper-rectangular partitions and predictions based on leaf nodes. In an ensemble, multiple trees are built from random training data samples, and their predictions are combined to improve accuracy.

1.4. LIME

Local Interpretable Model-agnostic Explanations (LIME) [21] offers a method for elucidating local predictions generated by black-box models. The process starts with selecting a data point of interest (DPoI) and then creating a perturbed dataset by independently sampling features. The black-box model is then applied to this perturbed dataset to produce labels, which serve as input for training a weighted Ridge Regression model. Perturbed points receive proximity-based weights through the Radial Basis Function (RBF) kernel. The regression coefficients provide insights into the behaviour of the black-box model around the DPoI, enhancing interpretability.

1.5. Research Objectives and Key Contributions

This study aims to develop highly interpretable local surrogate models using our novel LIMETree approach, which is an enhanced version of LIME, to explain DTE regression models for tabular data. The approach is applied to the PCC process data. The surrogate models will help investigate correlations among PCC process parameters by providing local explanations for DTE predictions. Ultimately, this will provide stable and measurable explanations that maintain high local fidelity to the DTE models.

The primary contribution of this research is the innovative LIMETree method, which addresses several limitations of the original LIME approach:

- LIMETree employs a Tree-Path proximity function to encapsulate the knowledge of the DTE models. This eliminates the assumption that black box models exhibit local linearity in their decision boundaries.
- It employs an L2 Hyper-Ellipsoid to define the local neighbourhood, improving the stability of the explanations and addressing the conflict between local fidelity and stability. This effectively resolves the neighbourhood vanishing issue associated with LIME's RBF kernel.
- By adding a new Tree core alongside LIME's default Ridge core, LIMETree provides a new form of explanation with higher local fidelity, interpretability, and flexibility.
- The Tree core also allows users to dynamically adjust its configuration to control the interpretability of model explanations. This flexibility supports a customizable approach to generate model predictions

according to individual preferences.

- The approach enhances the evaluation of the model's explanations by integrating QoE metrics: the Coefficient Stability Index (CSI) and Variables Stability Index (VSI) [22] alongside LIME's only fidelity metric, R^2 .
- Unlike LIME, which uses a simple estimate, LIMETree can identify the optimal kernel width through Bayesian optimization.

2. Methodology

This section focuses on applying LIMETree to the PCC process dataset.

2.1. Dataset

A domain expert utilized ProMax to create the training dataset. ProMax is a widely used software for seismic data processing that aids in analyzing and visualizing subsurface geological structures [22]. It enables geophysicists to process raw seismic data efficiently, offering valuable insights for oil and gas exploration and other geological applications. ProMax is also frequently employed to simulate the PCC process [23-25]. Furthermore, simulation data can be suitable for machine learning modelling if the data have been thoroughly validated by a domain expert [26]. When using simulation data along with machine learning, the simulation dataset can act as either supplemental training data [27] or as the primary training dataset for model development [28]. This work uses the ProMax simulation data as the sole training dataset.

During the simulation process, the specifications for the contactors were derived from existing CO₂ capture units in North America. The solvent and operating conditions were chosen based on the domain expert's assumptions. ProMax functions as a predictive tool that generates data. It was initially validated with acceptable deviations of less than 10%, which is a low prediction error for ProMax in an MEA operation based on mass and heat transfer methodology compared to experimental data from a pilot study [29].

The outputs or labels include two main parameters that indicate PCC process performance: Carbon Capture Efficiency (CCE) and Regeneration Energy. Carbon Capture Efficiency (CCE) measures the percentage of CO₂ removed from flue gas during capture. For instance, a 90% efficiency means 90% of CO₂ from the exhaust gas is captured. Apparently, higher CCE captures more CO₂, which aids climate change efforts. However, improving carbon capture efficiency often requires advanced and costly equipment. It also increases system complexity. Additionally, greater efficiency can increase operating costs due to higher energy demands and more processing steps. Thus, balancing cost and efficiency in carbon capture is crucial for maximizing environmental benefits without excessive cost. Regeneration energy refers to the energy needed to release captured CO₂ from the solvent for reuse during capture, and the process of releasing CO₂ occurs in the regeneration column's bottom reboiler. Since regeneration energy has a significant impact on operational costs, reducing regeneration energy lowers overall capture costs. A high energy requirement can also reduce the net efficiency of the carbon capture process. Therefore, a key goal in designing a PCC process is balancing high CCE with low regeneration energy, so as to ensure cost-effectiveness and minimal environmental impact of the carbon capture systems.

Ten PCC process parameters are key to CO₂ capture performance:

- Liquid Flow Rate: The amount of liquid passing through a specific area over time, affecting CO₂ contact and mass transfer. A higher flow rate improves absorption but can increase energy needs for pumping and regeneration. Excess flow may also increase reboiler duty and flooding risks.
- Lean Loading: CO₂ content in the solvent after regeneration. A lower lean loading improves CO₂ capture but may increase energy consumption for regeneration.
- Amine Concentration: A higher concentration enhances CO₂ absorption but may increase the energy demand for solvent regeneration and risks corrosion or degradation.

- **Reboiler Temperature:** This affects CO₂ desorption. A higher temperature generally improves CO₂ release but increases energy consumption, but overheating the solvent may cause degradation issues.
- **CO₂ Concentration in Flue Gas:** A higher concentration improves absorption efficiency but affects equipment size and capacity.
- **Rich Solvent Feed Temperature:** This is the temperature of the CO₂-rich solvent entering the regenerator. Optimizing this temperature lowers energy needs by reducing the temperature differences between the solvent and reboiler, which is related to the sensible heat.
- **Rich Loading:** This refers to the CO₂ content in the solvent after absorption. A higher rich loading indicates effective capture but requires more energy for solvent regeneration.
- **Flooding Factor:** This factor is directly impacted by the column's diameter and it indicates the risk of reduced mass transfer efficiency. Excessive flooding decreases CO₂ capture and increases energy use due to inefficiencies in the system.
- **Gas Flow Rate:** This is the speed at which flue gas enters the absorber. An excessively high flow rate reduces residence time, lowers capture efficiency, and increases energy costs.
- **Lean Solvent Temperature:** This is the temperature of the regenerated solvent before it re-enters the absorber. A lower temperature improves CO₂ capture but may increase cooling energy needs.

The value ranges and units of the columns of the dataset are shown in Table 1.

The dataset has only 200 data points, which is inadequate for effectively completing a DTE training task. Due to the limited dataset, the DTE model's accuracy is only 80%. A data synthesizer like the Conditional Tabular Generative Adversarial Network (CT-GAN) was adopted to tackle this problem because a CT-GAN can learn underlying patterns and distributions of the data and generate new, synthetic data similar to the original dataset. The CT-GAN

Table 1: Value ranges and units of the dataset

	Name	Range	Unit
Label	Carbon Capture Efficiency	[27.27, 99.75]	%
	Regeneration Energy	[0.01, 21.97]	$\frac{GJ}{ton}$
Feature	Liquid Flow Rate	[16.84, 80.00]	$\frac{m^3}{hour}$
	Lean loading	[0.10, 0.45]	$\frac{mol}{mol}$
	Amine concentration	[4.00, 7.50]	$\frac{mol}{mol}$
	Reboiler temperature	[112.41, 127.75]	$^{\circ}C$
	CO ₂ concentration on the gas side	[3.07, 14.43]	%
	Rich Solvent Feed Temperature	[80.00, 105.00]	$^{\circ}C$
	Rich loading	[0.33, 0.56]	$\frac{mol}{mol}$
	Flooding factor	[40.00, 95.00]	%
	Gas Flow Rate	[20326.00, 64597.00]	$\frac{m^3}{hour}$
	Lean solvent temperature after cooling	[20.00, 37.78]	$^{\circ}C$

synthesizer can utilize specific phenomena as conditionings to generate more realistic data [14, 30]. In this study, three phenomena identified during the preliminary analysis of the dataset were employed as constraints for the CT-GAN, which was used to synthesize 500 realistic data points. Domain experts meticulously reviewed the synthetic dataset to ensure its quality and authenticity before incorporating it into the final training dataset. Table 2 provides a detailed statistical comparison of the original and generated datasets. Based on the statistical review and feedback from domain experts, we concluded the CT-GAN generated high-quality synthetic data, and the data are suitable for our use.

2.2. Models

Two types of DTEs were used to build the ML models:

1. A group of standard models based on Breiman et al.'s work [19] was trained as a baseline. This type of model uses the classical tree-growing algorithm, so each node is split independently from other nodes. In other words, it will be split as long as a node fulfills the split constraints, such as "maximum depth," "maximum nodes," etc.

2. GBDT A group of GBDT [20] models was built for higher accuracy and simplicity. The simplicity of GBDT arises from their use of shallow DTs. The term "gradient boosting" [20] refers to the process where each decision tree is trained to predict and correct the errors made by the previously trained trees. Each tree predicts how the loss function should be adjusted to enhance the model's accuracy. Thus, each tree indicates the direction in which the overall combination of trees should move. The loss gradient provides insight into how the model should be modified to minimize the loss. Due to these characteristics, it is reasonable to expect that the gradient-boosting training algorithm will yield highly accurate predictions [20, 31]. When splitting the training set to grow a tree, Very Sparse Random

Projection [32] was utilized instead of the traditional axis-aligned splits commonly employed in standard DFs. The tree-growing method is based on Best-First Decision Tree Learning [33].

2.3. Explanations

Figure 1 shows sample explanations produced by the LIMETree methods. The top left section displays the feature values of the DPoI (Marker 1). Only three features are shown because LIMETree was requested to select the three most important features to ensure the explanations were easily comprehensible.

The prediction from the local surrogate model based on the DPoI is shown on the right side of the feature values (Marker 2). The top right corner shows a graphical representation of the surrogate model's coefficients (Marker 3). In this plot, the coefficients are colour-coded: orange for positive coefficients, indicating that higher feature values lead to higher predictions, and blue for negative coefficients, indicating that higher feature values result in lower predictions.

The bottom half of Figure 1 is the explanation offered by a Tree core for the same DPoIs. The Tree core can provide an explanation that reflects the significance of each feature. However, because the feature importance is determined through Permuted Feature Importance (PFI) analysis, the results cannot indicate whether each feature positively or negatively affects the prediction, as shown in the middle of the figure (Marker 4). For instance, in Figure 1, the feature importance plot (Marker 4) for the Tree core shows agreement on the dominant feature choice with the Ridge core plot (Marker 3) but disagreement on the others.

The Tree core visually illustrates the tree-based decision process of making predictions. It begins at the root and progresses downward to a leaf for the final prediction. As shown in Figure 1 (Marker 5), each non-leaf node displays the decision-making feature, the error from tree splitting, the available data points for tree growth, and the prediction at that node. Nodes with more saturated colours indicate it is on a decision path closer to the final prediction.

Table 3: An example of a two layers surrogate tree model's predicting rule for CCE

Condition	Prediction	Support(%)
((CO ₂ concentration on the gas side) > 5.67) and ((Rich Loading) ≤ 0.547)	56.873	31.40
((CO ₂ concentration on the gas side) ≤ 5.67) and ((Rich Loading) > 0.532)	82.736	26.20
((CO ₂ concentration on the gas side) ≤ 5.67) and ((Rich Loading) ≤ 0.532)	66.808	22.40
((CO ₂ concentration on the gas side) > 5.67) and ((Rich Loading) > 0.547)	66.534	20.00

showing the label. A vertical dashed line marks the decision boundary, while an orange triangle below the X-axis indicates the current feature's value. If the value is on the left side of the boundary, the path follows the left branch; if it is on the right, it follows the right branch. The orange path illustrates the decision pathway, with sample values at the bottom. For instance, as shown in a magnified view (Marker 7) in Figure 1, a prediction process involves two decisions based on "Rich Loading." The first decision branches to the left because "Rich Loading" ≤ 0.4989. The second decision branches to the right due to "Rich Loading" > 0.4761. Following these decisions, the process arrives at a leaf node that provides the final prediction of 75.47.

Table 3 illustrates the tree core's prediction rules, which provide an alternative explanation. Rather than obscuring the prediction's decision-making process, the rules guiding the tree model's predictions are presented. The example in Table 3 features a bare two-layer tree with relatively lower fidelity and high interpretability.

3. Results and Discussion

3.1. The Performance of DTE

To evaluate the performance improvement offered by CT-GAN and PFA when using DTEs as regressors for modelling tabular data, two DTE models were trained with 200 data points of the original dataset that included all

Another tree's statistical view is shown in Figure 1 (Marker 6). Each node features a plot of available data points, with the X-axis representing the current decision-making feature and the Y-axis

Table 5: DTE Models' performance comparison

Label	ID	Arch.	RMSE(%)	R ²	MAPE(%)
Carbon Capture Efficiency (%) [*]	1	GBDT	8.56	0.908	4.18
	2	GBDT	9.42	0.889	4.26
	3	GBDT	10.22	0.869	5.38
	4	GBDT	9.54	0.886	5.66
	5	GBDT	10.97	0.849	5.80
Carbon Capture Efficiency (%)	B1	DF	37.02	-0.720	22.39
	B2	GBDT	19.47	0.524	12.71
Regeneration Energy ($\frac{GJ}{ton}$) [*]	1	DF	11.58	0.709	8.77
	2	DF	11.48	0.714	8.95
	3	DF	11.94	0.691	9.36
	4	GBDT	9.94	0.786	9.43
	5	DF	11.74	0.701	9.46
Regeneration Energy ($\frac{GJ}{ton}$)	B1	DF	18.06	0.293	21.50
	B2	DF	13.46	0.607	19.13

Table 4: Performance Comparison Between Models in Groups B1 and B2

Model Group	Label	RMSE	R ²	MAPE
B1	Carbon Capture Efficiency (%)	22.650	-0.720	22.39%
B2	Carbon Capture Efficiency (%)	11.912	0.524	12.71%
B1	Regeneration Intensity ($\frac{GJ}{ton}$)	2.081	0.293	21.50%
B2	Regeneration Intensity ($\frac{GJ}{ton}$)	1.551	0.607	19.13%

9.68% for Root Mean Squared Error (RMSE), Coefficient of Determination(R-squared) (R²) and MAPE, respectively. In comparison, the “Regeneration Intensity” models achieved performance enhancements of 0.53, 0.324 and 2.37% for RMSE, R² and MAPE, respectively. In summary, the new models (Baseline model group B2) demonstrate improved performance compared to the B1 group baseline models.

To enhance the performance of models, we utilized the PFA to identify the most influential features before training the models. The top ten feature sets selected by various PFA configurations are displayed in Table 6, with IDs ranging from 1 to 5. Each row shows a feature set, where “” indicates a non-selected feature and “a” signifies a selected feature. In total, 20 DTE models were trained, including 10 DF and 10 GBDT models. Validation error was used to determine the best-performing models for each label, architecture and feature set combination. As indicated in Table 5, GBDT generally outperforms DF for the CCE label. However, for the label “Regeneration Energy,” the DF model tends to win in the majority of comparisons.

Comparing the baseline model group B1 to the highest-performing models (rows with * in Table 5), we observe the performance enhancements of CCE models are 28.46, 1.628 and 18.21% for RMSE, R² and MAPE, respectively, as demonstrated in Table 5. Similarly, for the label “Regeneration Energy,” there is also a significant performance gain of 6.48, 0.416 and 12.73% for RMSE, R² and MAPE, respectively. In conclusion, the performance improvements from the CT-GAN synthesizer and PFA feature selector are evident. Based on our preliminary testing results, we believe that the toolset — which includes CT-GAN, DTE, and PFA — proves highly effective in performing regression tasks for small tabular datasets with high dimensionality.

3.2. LIME vs LIMETree

We designed a group of experiments to compare the Quality of Explanation (QoE) differences between LIME and variations of the LIMETree configurations. The experiments are explained next.

features as the baseline models. This is called the Baseline model group B1. The performance of the models, as shown in Table 4 and Figure 2, was unsatisfactory. Both models exhibit a Mean Absolute Percentage Error (MAPE) higher than 20% and an R² < 0.3. This is because a limited training set can lead to difficulties in constructing a DTE [19]. To address this issue, we used CT-GAN to generate 500 additional data points for the training set. The 500 synthesized data points are merged with 162 real data points to create a new training set, and 32 real data points are reserved for a testing set. With 662 data points that include all features, four models are trained: 2 for DF and 2 for Stochastic Gradient-Boosted Decision Trees (GBDT). Validation error was used to determine the best-performing model for each combination of label and architecture. As demonstrated in Table 4, the performance enhancements that CT-GAN provides to CCE models are 10.738, 1.244 and

Firstly, a domain expert selected ten data points from the original Promax dataset to represent the most significant system states, we called this the LIME testing set. Next, black-box models are selected for the experiments. As indicated in Tables 5 and 6, two models belonging to the B1 group were removed from the experiment due to their low performance. Therefore, the other twelve models were used as black-box models.

Table 6: Decision Tree Ensembles (DTEs) models' feature sets. CCE stands for Carbon Capture Efficiency, and RE stands for Regeneration Energy

Label	ID	Liquid Flow Rate	Lean Loading	Amine Concentration	Reboiler Temperature	CO ₂ Concentration on Gas Side	Rich Solvent Feed Temperature	Rich Loading	Flooding Factor	Gas Flow Rate	Lean Solvent Temperature After Cooling
		$\frac{m^3}{hour}$	$\frac{mol}{mol}$	$\frac{mol}{L}$	$^{\circ}C$	%	$^{\circ}C$	$\frac{mol}{mol}$	%	$\frac{m^3}{hour}$	$^{\circ}C$
CCE	1	✓	×	✓	✓	✓	×	✓	×	×	×
CCE	2	✓	✓	✓	✓	✓	×	✓	×	×	×
CCE	3	✓	×	✓	✓	✓	×	✓	×	✓	×
CCE	4	✓	✓	✓	✓	✓	×	✓	×	✓	×
CCE	5	✓	✓	×	✓	✓	×	✓	✓	×	×
CCE	B1	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
CCE	B2	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
RE	1	✓	✓	✓	✓	✓	✓	✓	✓	×	×
RE	2	✓	✓	✓	✓	✓	✓	✓	✓	×	×
RE	3	✓	✓	✓	✓	×	✓	✓	✓	×	✓
RE	4	✓	✓	✓	✓	×	×	✓	×	×	×
RE	5	✓	✓	✓	✓	×	✓	✓	✓	×	×
RE	B1	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
RE	B2	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

LIME's default kernel width of 0.75 times the number of features to ensure that experiments of LIME and LIMETree are comparable. Lastly, in the LIMETree experiments, each DPoI is explained by the LIMETree method, which was done by applying the LIMETree method with eight configurations regarding LIMETree's three key parameters of Core, Proximity, and Locality.

Figure 3 directly compares the Coefficient Stability Index (CSI) and R^2 to highlight the differences between the two methods. As illustrated in Figure 3, LIME's CSI ranges from 0 to 30, while LIMETree scores above 90. Furthermore, LIMETree's R^2 value is at least three times higher than LIME's. Therefore, we can conclude that LIMETree significantly outperforms LIME.

The Variables Stability Index (VSI) values are not shown in Figure 3 because they remain constant at 100. This consistency in VSI occurs because, in the experiments, both LIME and LIMETree utilized all available features when explaining the DPoIs rather than selecting the top k features, so the process inside LIME and LIMETree was turned off. Consequently, every feature included in each black box model was consistently available for selection for each explanation. Turning off the feature selection process allows both LIME and LIMETree to maximize their QoE, which aligns with the experiment's objectives.

3.3. A low-depth tree core with hyper-ellipsoid gives the best explanation

A hyper-ellipsoid emphasizes more important features along its longer axis. It happens because the hyper-ellipsoid determines the length of each semi-axis (radii) based on the relative importance of the features. As a result, the local surrogate model achieves higher fidelity. When generating perturbed datasets, the orientation of important features is crucial for accurately representing and effectively utilizing the generated data. If a key feature coincides with the longer axis of the data distribution in the altered space, it significantly boosts the surrogate model's capacity to mimic and examine the perturbed dataset.

To understand this concept, let us consider the geometry of data distribution. In multidimensional data, each feature can be visualized as a dimension in a coordinate space. The variance along these dimensions determines how spread out the data points are. Therefore, features that align with the directions of maximum variance are particularly important. When positioned along the longer axes, these significant features can capture more underlying patterns and structures in the dataset, indicating the direction of greater variance [15].

To elaborate on this concept, we can see that the integration of Hyper-Ellipsoid and Tree-Path forms a belt-like structure, illustrated in the left plot of Figure 4. In comparison, the RBF kernel constructs a mushroom-shaped data space, depicted on the right. This phenomenon occurs when a dominant feature is present in this application; that feature is “Rich Loading.” The dominant feature is significantly more important than others, leading to a perturbed

Table 7: An example of an explanation

When Rich Loading ≤ 0.512 , the surrogate model predicts CCE as 79.999, supported by 71.80% of samples.
Otherwise, it predicts CCE as 57.645, supported by 28.20% of samples.
$R^2 = 86.9 \pm 1.3$, CSI = 100 and VSI = 92.79.

space that resembles a belt shape. The primary advantage of a belt-like perturbed data space is its ability to provide clear and interpretable explanations while

maintaining high stability and fidelity. For example, the left plot in Figure 4 presents the explanation with high interpretability, and the details are shown in Table 7.

4. Conclusion and Future Work

Our work proposes a novel approach called Tree-Based Local Interpretable Model-agnostic Explanation (LIMETree). This approach generates highly interpretable local surrogate models to explain the predictions made by the Decision Tree Ensembles (DTE) model in regression tasks involving tabular data. This approach supports examining feature correlations within the tabular dataset by providing stable and measurable explanations for the predictions of a black box model within a local context. These explanations help stakeholders better understand machine learning models and their predictions.

LIMETree addressed several limitations of the original LIME framework. While LIME primarily relies on a local fidelity metric, the R^2 , LIMETree enhances this by incorporating the CSI and VSI into its QoE metrics for evaluating the explanations. For the crucial parameter of kernel width, which defines the size of the local neighbourhood, LIMETree employs Bayesian Optimization to determine the optimal value based on the adjusted R^2 . In contrast, LIME typically sets it at 0.75 times the square root of the number of features. Additionally, LIMETree supports the Tree-Path distance function, which utilizes proximity knowledge from the DTE rather than relying solely on the Euclidean distance of LIME. To address the neighbourhood vanishing problem associated with the RBF in LIME, LIMETree incorporates the Hyper-Ellipsoid method. This method not only solved the neighbourhood vanishing problem but also provided a high-quality perturbed data space by placing greater emphasis on the most influential features. Finally, LIMETree integrates the decision tree core, which enhances the richness of the explanation forms and results in a higher fidelity surrogate model.

The LIMETree has been applied to a Promax-generated PCC process dataset, where the specifications of contactors are based on existing CO₂ capture units in North America. The preliminary results clearly demonstrate that our method is useful for developing interpretable and more comprehensive machine-learning models. We shall continue our efforts to ensure these models have a meaningful impact on critical real-world missions.

Some directions for future research include:

- We will apply LIMETree to additional engineering datasets to confirm its effectiveness in generating interpretable ML models.
- After the algorithm demonstrates satisfactory performance on both experimental and engineering datasets, we aim to develop it into a decision support system so that more non-ML researchers can utilize it for data analysis.

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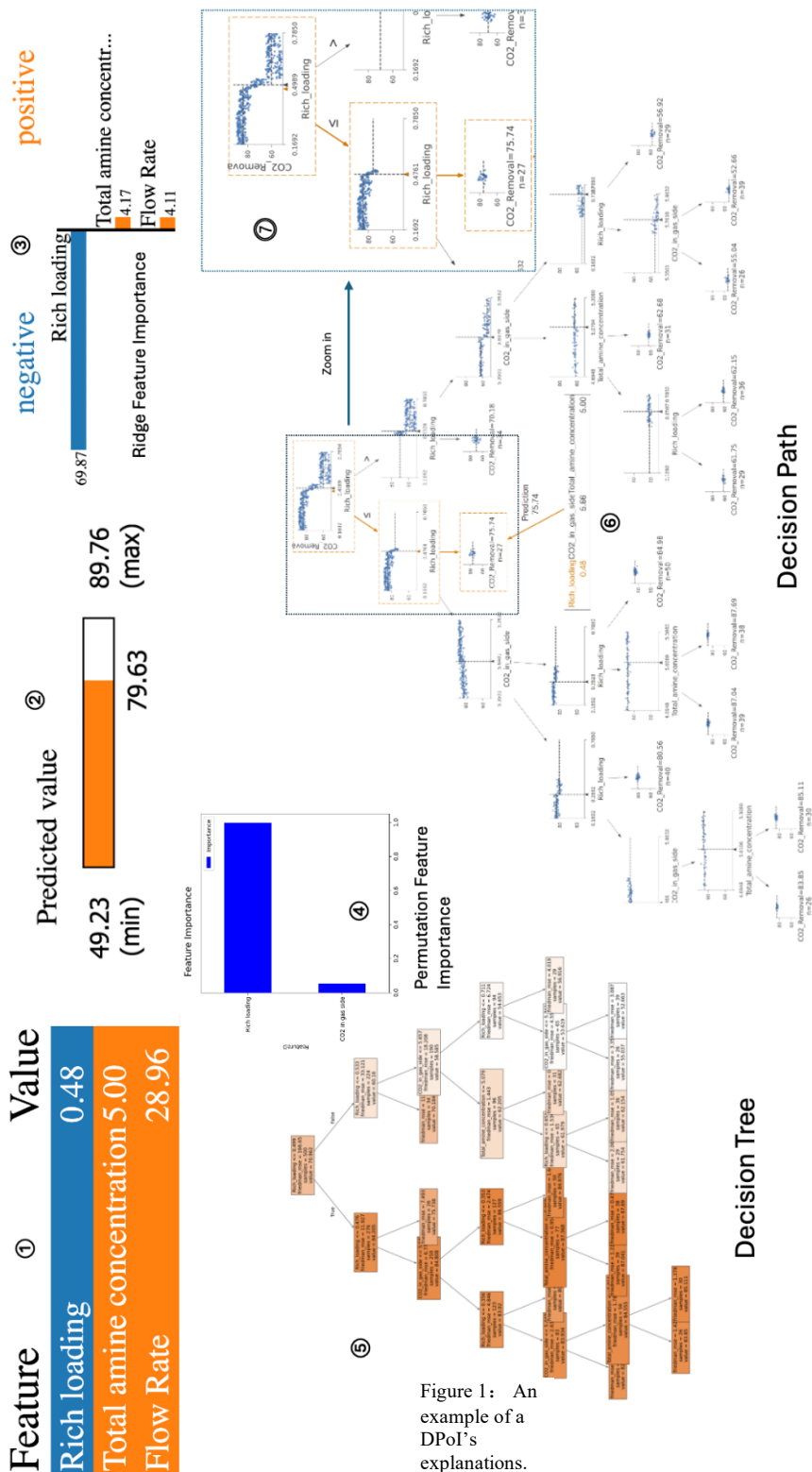


Figure 1: An example of a DPoI's explanations.

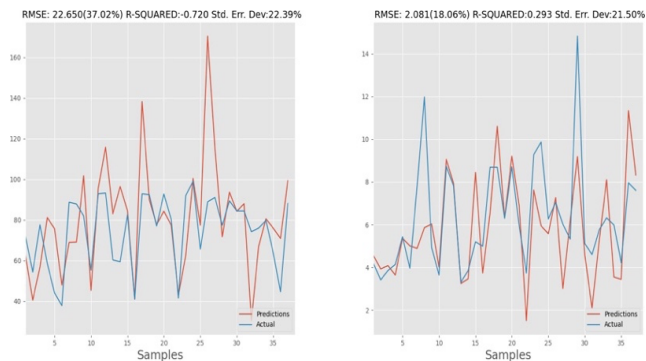


Figure 2: The performance of the DF models with the original dataset

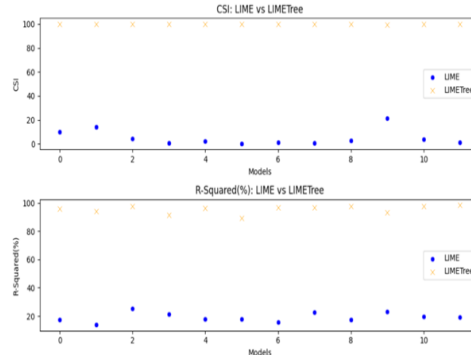


Figure 3: QoE: LIME vs LIMETree

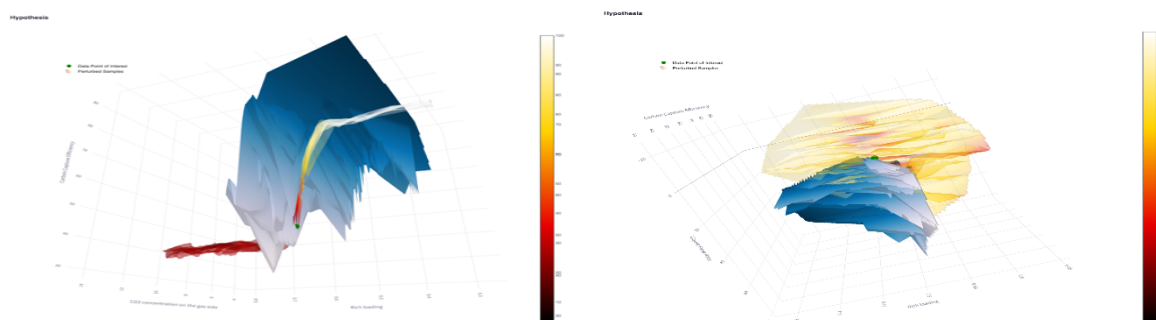


Figure 4: The perturbed data samples: Hyper-Ellipsoid vs RBF Kernel

Table 2: Statistical comparison between synthetic and real data. The upper section contains real data, while the lower section features synthetic data.

Feature	mean	std	min	25%	50%	75%	max
Carbon Capture Efficiency (%)	72.52	17.50	27.27	61.10	75.40	85.29	99.75
Regeneration Energy ($\frac{GJ}{ton}$)	6.60	3.65	0.01	4.23	5.38	7.76	21.97
Liquid Flow Rate ($\frac{m^3}{hour}$)	40.92	15.13	16.84	30.00	38.92	50.00	80.00
Lean loading ($\frac{mol}{mol}$)	0.26	0.09	0.10	0.18	0.26	0.32	0.45
Amine concentration ($\frac{mol}{L}$)	5.79	1.35	4.00	5.00	5.00	7.50	7.50
Reboiler temperature ($^{\circ}C$)	122.76	3.30	112.41	121.71	124.19	124.97	127.75
CO_2 concentration on the gas side (%)	7.91	3.11	3.07	5.66	5.66	10.15	14.43
Rich Solvent Feed Temperature ($^{\circ}C$)	97.69	9.36	80.00	90.00	105.00	105.00	105.00
Rich loading ($\frac{mol}{mol}$)	0.48	0.05	0.33	0.46	0.50	0.52	0.56
Flooding factor (%)	73.95	12.45	40.00	65.00	75.00	85.00	95.00
Gas flow rate ($\frac{m^3}{hour}$)	42371.89	14722.06	20326.00	32068.50	50859.00	50859.00	64597.00
Lean solvent temperature after cooling ($^{\circ}C$)	35.40	5.53	20.00	37.78	37.78	37.78	37.78
Carbon Capture Efficiency (%)	77.16	16.36	27.27	64.40	80.90	90.10	99.75
Regeneration Energy ($\frac{GJ}{ton}$)	6.14	3.04	0.01	4.12	5.00	7.11	21.97
Liquid Flow Rate ($\frac{m^3}{hour}$)	40.26	12.24	16.84	31.91	39.24	47.35	75.97
Lean loading ($\frac{mol}{mol}$)	0.26	0.09	0.10	0.18	0.26	0.33	0.45
Amine concentration ($\frac{mol}{L}$)	5.90	1.31	4.00	4.93	5.15	7.25	7.50
Reboiler temperature ($^{\circ}C$)	123.09	3.12	112.41	122.36	123.99	124.94	127.75
CO_2 concentration on the gas side (%)	7.61	2.96	3.66	5.58	5.71	10.53	14.43
Rich Solvent Feed Temperature ($^{\circ}C$)	94.96	9.54	80.00	90.00	100.00	105.00	105.00
Rich loading ($\frac{mol}{mol}$)	0.48	0.05	0.33	0.46	0.49	0.51	0.56
Flooding factor (%)	71.37	13.24	40.00	62.00	70.00	84.00	95.00
Gas flow rate ($\frac{m^3}{hour}$)	42197.54	13503.50	20326.00	28008.17	50218.65	51370.67	64597.00
Lean solvent temperature after cooling ($^{\circ}C$)	35.63	4.47	20.59	37.61	37.78	37.78	37.78

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